

Aylar Babaei

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PROFESSIONAL SUMMARY

Bioinformatics Scientist with experience in genomics and metabolomics, including building and running production pipelines for pediatric cancer research. Writes and maintains Python, R, and Bash code on Linux, AWS, and CAVATICA; harmonizes RNA-seq and DNA sequencing data across multiple cohorts; and built the HLA typing and neoantigen prediction workflows used in the lab's immunotherapy-focused research. Works closely with clinicians and bench scientists on study design, analysis, and interpretation. Develops and automates pipelines and data infrastructure.

CORE SKILLS

Programming & Analysis: Python, R, Bash/Shell, SQL; statistical analysis and visualization of genomic, transcriptomic, and metabolomic data

Machine Learning: Random forests and tree ensembles, regularized regression (LASSO, ridge), neural networks; scikit-learn, R (caret, tidymodels, randomForest)

Pipelines: CWL, Docker; building, testing, and debugging production pipelines; QC automation; validation and benchmarking

Cloud & Compute: AWS (S3, EC2), CAVATICA/Seven Bridges, HPC (VM provisioning, job scheduling), Linux; performance profiling and optimization

Version Control: Git/GitHub (pull requests, code review); SOP documentation

Data Operations: Metadata harmonization, FAIR data practices, manifest generation, schema validation, relational table structures

Domain: RNA-seq, WGS/WES, DNA methylation, HLA typing, neoantigen prediction; GATK, gnomAD, ClinVar; FASTQ, BAM, CRAM, VCF, MAF, IDAT

EXPERIENCE

Bioinformatics Scientist

Children's National Hospital, BTI Bioinformatics Core · Washington, DC | Jun 2024 – Present

- Developed and improved production pipelines for multi-cohort genomic data harmonization, including building a mouse RNA-seq workflow from raw reads through quantification and QC; debug pipeline errors, rerun failed jobs, and document fixes to support reproducibility.
- Built HLA typing and neoantigen prediction workflows (RNA2HLA, arcashLA, IEDB-based MHC I binding) and contributed to noncanonical antigen discovery with Ribo-seq for immunogenomics studies.
- Automated recurring analysis and data-processing workflows, reducing manual steps across lab projects.
- Contributed to building a data warehouse integrating genomic data and sample metadata across cohorts, supporting centralized querying and downstream analysis.
- Identified and flagged data errors in clinical trial datasets through systematic QC and validation, improving data integrity before analysis.
- Run pipelines on AWS (S3, EC2) and CAVATICA/Seven Bridges; provision HPC/VM compute, manage job scheduling, and tune runtime, memory, and resource allocation.
- Contribute to production pipeline pull requests via Git/GitHub, including methylation-based rare tumor subtyping within a national genomics consortium.

Graduate Research Assistant

GWU Computational Biology Institute, Rahnavard Lab · Washington, DC | Jan 2023 – May 2024

- Profiled and quantified LC/MS metabolomics data from iHMP studies to support development of the massSight software package, including quality assessment, review of the package's machine learning model outputs, and biological interpretation of high-dimensional datasets.

Graduate Lab Assistant

GWU Cancer Center · Washington, DC | Mar 2023 – Dec 2023

- Supported translational cancer biology studies on CDKN2A/P16 deletion and KRAS activation in mouse intestinal epithelial cells; assisted with Visium spatial transcriptomics generation, IHC workflows, and genotyping.
- Managed a database of 2,000+ laboratory mice, ensuring data accuracy and integrity across breeding, genotype, and health records.

HONORS & PROJECTS

St. Jude BioHackathon 2026 – Team 11 | 2026

- Awarded a competitive travel award to participate in the St. Jude Children's Research Hospital BioHackathon.
- Co-developed "FAIR Harmonization and Automated LabKey Integration of Legacy Sequencing Metadata," cleaning and unifying legacy sequencing metadata from physical lab notes into a FAIR-compliant schema for automated LabKey ingestion (github.com/stjude-biohackathon/KIDS26-Team11).

SELECTED PUBLICATIONS

- Large-Scale Assessment of NF1 Single Amino Acid Variants as HLA Class I Neoantigens. *bioRxiv*, 2026.
- Peer-reviewed article. *Cells*, 2025.
- Abstract. *Neuro-Oncology (Pediatrics)*, 2026.
- Abstract. *Gastroenterology*, 2025.

EDUCATION

M.S. Bioinformatics, The George Washington University | Aug 2022 – May 2024

B.S. Chemistry, Shiraz University | Sep 2014 – May 2018